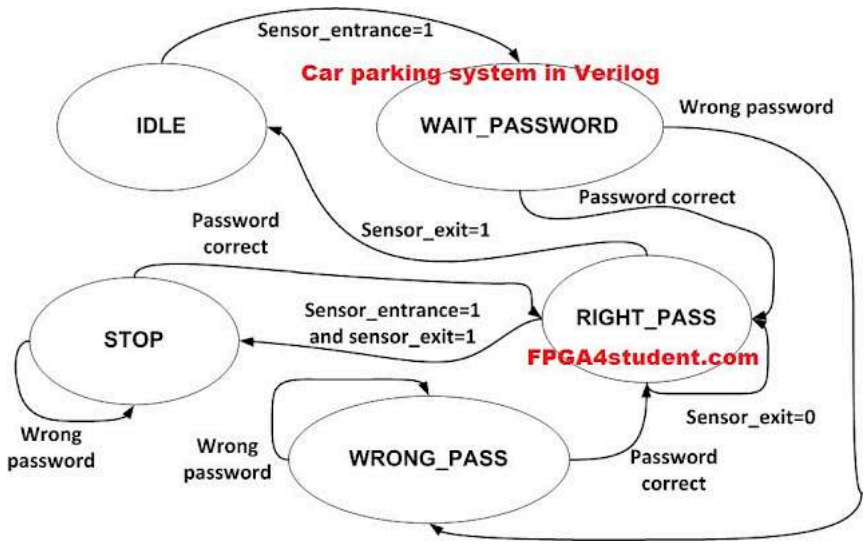
Verilog code for Car Parking System

This simple project is to implement a car parking system in Verilog. The Verilog code for the car parking system is fully presented.

In the entrance of the parking system, there is a sensor which is activated to detect a vehicle coming. Once the sensor is triggered, a password is requested to open the gate. If the entered password is correct, the gate would open to let the vehicle get in. Otherwise, the gate is still locked. If the current car is getting in the car park being detected by the exit sensor and another the car comes, the door will be locked and requires the coming car to enter passwords.



Verilog code for the car parking system:

```
// fpga4student.com FPGA projects, Verilog projects, VHDL projects
// Verilog project: Verilog code for car parking system
`timescale 1ns / 1ps
module parking_system(
                input clk,reset_n,
 input sensor_entrance, sensor_exit,
 input [1:0] password_1, password_2,
 output wire GREEN_LED,RED_LED,
 output reg [6:0] HEX_1, HEX_2
    );
 parameter IDLE = 3'b000, WAIT_PASSWORD = 3'b001, WRONG_PASS = 3'b010, RIGHT_PASS
 // Moore FSM : output just depends on the current state
 reg[2:0] current_state, next_state;
 reg[31:0] counter_wait;
 reg red_tmp,green_tmp;
 // Next state
 always @(posedge clk or negedge reset_n)
 begin
 if(~reset_n)
 current_state = IDLE;
 else
 current_state = next_state;
 end
 // counter_wait
 always @(posedge clk or negedge reset_n)
 begin
 if(~reset_n)
 counter_wait <= 0;
 else if(current_state==WAIT_PASSWORD)
 counter_wait <= counter_wait + 1;
```

```
 else
 counter_wait <= 0;
 end
 // change state
// fpga4student.com FPGA projects, Verilog projects, VHDL projects
 always @(*)
 begin
 case(current_state)
 IDLE: begin
        if(sensor_entrance == 1)
 next_state = WAIT_PASSWORD;
 else
 next_state = IDLE;
 end
 WAIT_PASSWORD: begin
 if(counter_wait <= 3)
 next_state = WAIT_PASSWORD;
 else
 begin
 if((password_1==2'b01)&&(password_2==2'b10))
 next_state = RIGHT_PASS;
 else
 next_state = WRONG_PASS;
 end
 end
 WRONG_PASS: begin
 if((password_1==2'b01)&&(password_2==2'b10))
 next_state = RIGHT_PASS;
 else
 next_state = WRONG_PASS;
 end
 RIGHT_PASS: begin
 if(sensor_entrance==1 && sensor_exit == 1)
 next_state = STOP;
 else if(sensor_exit == 1)
 next_state = IDLE;
 else
 next_state = RIGHT_PASS;
 end
 STOP: begin
 if((password_1==2'b01)&&(password_2==2'b10))
 next_state = RIGHT_PASS;
 else
 next_state = STOP;
 end
 default: next_state = IDLE;
 endcase
 end
 // LEDs and output, change the period of blinking LEDs here
 always @(posedge clk) begin
 case(current_state)
 IDLE: begin
 green_tmp = 1'b0;
 red_tmp = 1'b0;
 HEX_1 = 7'b1111111; // off
 HEX_2 = 7'b1111111; // off
 end
 WAIT_PASSWORD: begin
 green_tmp = 1'b0;
 red_tmp = 1'b1;
 HEX_1 = 7'b000_0110; // E
 HEX_2 = 7'b010_1011; // n
 end
```

```
 WRONG_PASS: begin
 green_tmp = 1'b0;
 red_tmp = ~red_tmp;
 HEX_1 = 7'b000_0110; // E
 HEX_2 = 7'b000_0110; // E
 end
 RIGHT_PASS: begin
 green_tmp = ~green_tmp;
 red_tmp = 1'b0;
 HEX_1 = 7'b000_0010; // 6
 HEX_2 = 7'b100_0000; // 0
 end
 STOP: begin
 green_tmp = 1'b0;
 red_tmp = ~red_tmp;
 HEX_1 = 7'b001_0010; // 5
 HEX_2 = 7'b000_1100; // P
 end
 endcase
 end
 assign RED_LED = red_tmp  ;
 assign GREEN_LED = green_tmp;

endmodule
```

Testbench Verilog code for car parking system:

```
`timescale 1ns / 1ps
// fpga4student.com FPGA projects, Verilog projects, VHDL projects
// Verilog project: Verilog code for car parking system
module tb_parking_system;

  // Inputs
  reg clk;
  reg reset_n;
  reg sensor_entrance;
  reg sensor_exit;
  reg [1:0] password_1;
  reg [1:0] password_2;

  // Outputs
  wire GREEN_LED;
  wire RED_LED;
  wire [6:0] HEX_1;
  wire [6:0] HEX_2;
// fpga4student.com FPGA projects, Verilog projects, VHDL projects
  // Instantiate the Unit Under Test (UUT)
  parking_system uut (
  .clk(clk),
  .reset_n(reset_n),
  .sensor_entrance(sensor_entrance),
  .sensor_exit(sensor_exit),
  .password_1(password_1),
  .password_2(password_2),
  .GREEN_LED(GREEN_LED),
  .RED_LED(RED_LED),
  .HEX_1(HEX_1),
 .HEX_2(HEX_2)
 );
 initial begin
 clk = 0;
 forever #10 clk = ~clk;
 end
```

```
 initial begin
 // Initialize Inputs
 reset_n = 0;
 sensor_entrance = 0;
 sensor_exit = 0;
 password_1 = 0;
 password_2 = 0;
 // Wait 100 ns for global reset to finish
 #100;
      reset_n = 1;
 #20;
 sensor_entrance = 1;
 #1000;
 sensor_entrance = 0;
 password_1 = 1;
 password_2 = 2;
 #2000;
 sensor_exit =1;

 // Add stimulus here
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 end

endmodule
```
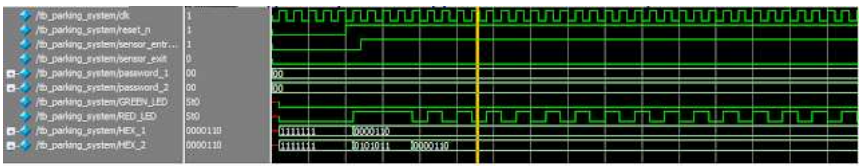

Simulation waveform for the car parking system in Verilog:



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